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Single LED visible light positioning scheme based on intelligent reflecting surfaces

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Visible light positioning (VLP) systems have gained attention for their ability to operate without additional infrastructure. However, most positioning schemes rely on multiple LEDs, which presents a considerable challenge in scenarios where only a single-light-source is available. To address this issue, this work proposes using intelligent reflective surfaces (IRS) as virtual light sources for VLP. To our knowledge, a novel positioning scheme, referred to as the difference of received signal strength, is proposed to mitigate spatial interference from multiple IRS and accurately estimate their power, enabling positioning even with a blocked line-of-sight link. Unlike maximum likelihood estimators, this scheme eliminates the need for a precise noise model. Meanwhile, the Cramér–Rao lower bound is derived to analyze the impact of IRS parameters such as rotation angle, layout, and quantity, on positioning accuracy. Theoretical analysis indicates that IRS orientation and layout greatly influence positioning accuracy, with layout offering more significant improvement than IRS quantity. Simulation results demonstrate that the scheme achieves a positioning accuracy of 0.056 m with 16 IRS units, with further improvements through additional optimization. Moreover, the results align with theoretical analysis, with an optimized layout reducing the positioning error to 0.028 m, highlighting its pivotal role in enhancing IRS performance.

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1. INTRODUCTION

Positioning technology serves as a cornerstone of modern information society, providing precise location information that underpins advancements in navigation, logistics management, smart device interaction, and emergency response systems [1]. However, traditional radio frequency (RF) positioning technologies, such as GPS, Wi-Fi, and Bluetooth, face significant limitations in indoor environments due to signal attenuation and multipath effects, making it challenging to achieve high-accuracy localization [2]. Against this backdrop, visible light positioning (VLP) has emerged as a promising alternative. With its advantages of high precision, strong resistance to interference, energy efficiency of light-emitting diode (LED), and integration with lighting functionality, VLP has garnered increasing attention from both academia and industry [3].

Most current VLP systems rely on multi-light-source layouts and employ multiplexing techniques, such as time-division multiple access [4], frequency-division multiple access [5], or code-division multiple access [6] to mitigate signal interference between light sources. However, these systems often face challenges, such as high hardware complexity, elevated costs, and positioning accuracy affected by the multi-light-source layout. Furthermore, in scenarios, such as corridors, tunnels, and factories, where lights are typically arranged in a linear configuration with large spacing, user devices may be unable to receive signals from multiple light sources simultaneously, thereby limiting the applicability of multi-light-source systems. In contrast, single-light-source positioning systems offer significant advantages, including immunity to multi-source interference, reduced layout constraints, and improved energy efficiency, making them more suitable for small indoor spaces. Consequently, indoor positioning based on a single-light-source has become a focus of research.

In terms of VLP with a single-light-source, VLP approaches are primarily categorized into image sensor-based and photodetector (PD)-based approaches. In Ref. [7], an image sensor is employed to receive position information transmitted via visible light communication (VLC) and designs a positioning algorithm that is robust to receiver tilting for estimating the receiver's location. However, image sensor-based methods are difficult to meet the stringent requirements of high real-time performance and low power consumption of the Industrial Internet due to their high computational complexity (real-time image processing is required), low response rate (frame rate is limited), and large-size hardware structure (image sensor volume limitation) [7]. As a result, research in single-light-source positioning has increasingly shifted toward PD-based methods. In Ref. [2], a receiver comprising one horizontal PD and two tilted PDs is proposed by utilizing their relative positions and received signal strength (RSS), achieving centimeter-level positioning accuracy. Meanwhile, an array of tilted complementary photodiodes is designed to improve the field of view of the angle-of-arrival estimator [8]. By deriving a closed-form expression for the optimal orientation with any quantity of PDs in the array, the system achieves a positioning accuracy of 2 cm in single-light-source scenarios. However, these methods must obtain the rotation matrix of the receiver relative to the world coordinate system in advance, so additional auxiliary equipment is needed to measure the attitude of the receiver. To overcome this limitation, a novel positioning approach using a rotatable PD is introduced in a single-light-source scenario [9]. During the PD rotation process, the rotation angles corresponding to the maximum and minimum received power are identified. The azimuth angle is determined based on these two angles, while the elevation angle is derived from the received power, enabling the positioning of a single light source [9]. However, the rotating machine structure makes the system bulky, the signal is prone to fluctuations due to the PD angle sensitivity, and the dynamic response speed is limited, which is not conducive to positioning. In Ref. [10], this method first measures an RSS fingerprint database in the presence of indoor reflections and applies machine learning algorithms for localization. Simulation results demonstrate an average positioning accuracy of 1.74 cm. Consequently, the Cramér–Rao lower bound (CRLB) for the rotatable PD-based method is further investigated, optimizing the selection of PD rotation angles and combining least-squares estimation with an improved grey wolf optimization algorithm to enhance positioning accuracy. However, the above single-light-source positioning methods essentially necessitate modifications to the receiver to enable positioning, thereby increasing hardware cost and complexity.

In recent years, intelligent reflecting surfaces (IRS) have emerged as a promising technology in wireless communication systems due to their low energy consumption, ease of deployment, and programmability [11]. By leveraging a large quantity of passive reflective elements, IRS can modulate incident electromagnetic or optical waves and reflect them as new beams. This enables the construction of intelligent and reconfigurable virtual line-of-sight (LOS) links, enhancing signal reception power and coverage. As research progresses, the potential of IRS in the field of wireless localization has gained increasing attention [12]. By precisely controlling signal propagation paths, IRS not only mitigates issues, such as signal blockage and attenuation in complex environments but also improves the accuracy and robustness of localization systems, offering new perspectives and methodologies for advancing wireless positioning technologies. Studies have demonstrated the significant potential of IRS in wireless localization, especially for enabling user positioning in scenarios with insufficient transmitters [13]. For example, a device-to-device localization algorithm in millimeter-wave RF systems is proposed, which operates effectively even in the absence of access points [14]. In Ref. [15], the performance of IRS-assisted mmWave localization systems is analyzed, with CRLB derived and reflective beamforming optimized to improve accuracy. Additionally, a low-complexity fingerprinting-based localization algorithm is devised for wireless communication systems with IRS [16]. In Ref. [17], the localization and orientation performance of synchronous and asynchronous signaling schemes are analyzed by deriving the CRLB, and a closed-form IRS phase configuration scheme is proposed for joint communication and positioning. In Ref. [18], a general signal model for wideband systems with IRS is established, and the CRLB is derived to evaluate the positioning performance of IRS-assisted wideband systems. While IRS technology has been extensively studied in RF-based localization systems, its application in VLP remains underexplored, with only three studies reported to date. In Ref. [19], a sparse Bayesian learning algorithm is developed to address the localization problem in IRS-assisted indoor VLP. In Ref. [20], a position estimation problem based on RSS is formulated for IRS-assisted VLP, where a maximum likelihood (ML) estimator is derived using signals from both LOS and IRS components, accompanied by the corresponding CRLB. An optimization algorithm is further proposed to adjust the IRS elements for maximizing received power. In Ref. [21], the impact of IRS orientation mismatches on localization performance is examined by deriving the misspecified CRLB and the mismatched ML estimator. However, research on IRS-assisted VLP remains limited, with existing positioning methods primarily based on ML estimation, especially in single-light-source scenarios. These ML estimates require very accurate noise probability distribution functions. Therefore, further investigation is required to address the challenges posed by single-light-source IRS-aided VLP systems.

In this paper, IRS are implemented as a virtual light source to achieve single-light-source VLP. To mitigate spatial interference and enable robust positioning without relying on the probability distribution of received optical power, this work proposes a novel difference of received signal strength (DRSS) scheme to ensure reliable positioning in the absence of LOS. This work derives the CRLB to evaluate the effects of IRS parameters, including rotation angle, layout, and quantity, on positioning accuracy. The simulation results confirm the effectiveness of the proposed scheme, achieving a positioning accuracy of 0.056 m with only 16 IRS units under a noise level of 110 dB, with further improvements possible through parameter optimization. Moreover, compared to other parameters, optimizing the IRS layout has a more pronounced impact on positioning accuracy. Under identical conditions, an optimized layout reduces the positioning error from 0.056 to 0.028 m, highlighting the critical role of a well-designed layout in maximizing IRS performance.

2. OPERATION PRINCIPLE

A. IRS-aided VLP System Model with a Single LED

Figure 1 illustrates the proposed VLP system and its established world coordinate framework. The scenario involves a narrow corridor, where, to minimize interference between light sources, the distance between adjacent sources is relatively large, and they are evenly spaced. The direct illumination coverage is

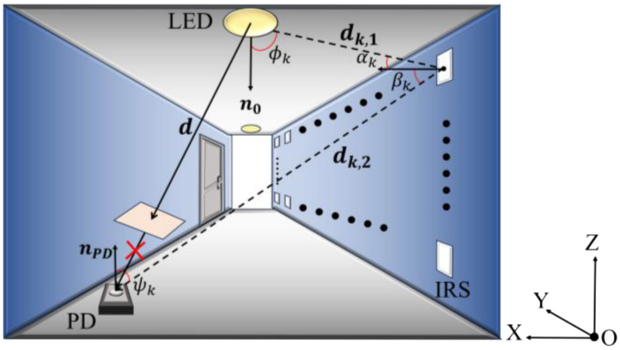

Fig. 1. Indoor visible light system structure.

limited, and the LOS link may be obstructed. To address these challenges, IRS are deployed on one or both walls of the corridor to assist the light sources in achieving accurate positioning. For the purposes of subsequent analysis, the corridor can be treated as a room with equal-sized segments and only two walls. The light source is located at $P_{\text{led}} = (x_0, y_0, z_0)$, the k_{th} IRS are positioned at $P_k = (x_k, y_k, z_k)$, and the PD is situated at $P_{\text{PD}} = (x, y, z)$. Let $\boldsymbol{d}$ represent the LOS link between the LED and the PD. The propagation path from the LED to the k_{th} IRS is defined as $\boldsymbol{d}_{\boldsymbol{k},1} = P_k - P_{\text{led}}$, while the path from the IRS to the PD is expressed as $\boldsymbol{d}_{\boldsymbol{k},2} = P_{\text{PD}} - P_k$. The orientation vectors of the LED, IRS, and PD are denoted by $\boldsymbol{n}_0$, $\boldsymbol{n_k}$, and $\boldsymbol{n_{\text{PD}}}$, respectively, with $\boldsymbol{n}_0$ defaulting to vertically downward, $\boldsymbol{n_k}$ horizontally inward, and $\boldsymbol{n_{\text{PD}}}$ vertically upward.

Based on the Lambert model, the received power can be obtained as follows:

$$P_{\text{RX}}(\boldsymbol{v}) = P_{\text{LOS}}(\boldsymbol{v}) + \sum_{k=1}^{N} P_{\text{IRS},k}(\boldsymbol{v}) + \eta, \tag{1}$$

where $\boldsymbol{v}$ is the PD coordinate vector, i.e., $\boldsymbol{v} = (x, y, z)$. $P_{\text{RX}}(\boldsymbol{x})$ represents the power received by the PD, $P_{\text{LOS}}(\boldsymbol{v})$ is the power of the LOS, $P_{\text{IRS},k}(\boldsymbol{v})$ is the power reflected from the k_{th} IRS to the PD, N is the quantity of IRS, and η is the zero-mean Gaussian noise with a variance of σ^2. The expressions of $P_{\text{LOS}}(\boldsymbol{v})$ and $P_{\text{IRS},k}(\boldsymbol{v})$ are

$$P_{\text{LOS}}(\boldsymbol{v}) = \frac{P_0(m_k+1)A_{\text{PD}}(\cos\phi_k)^{m_k} T_s(\psi_k) g(\psi_k)\cos\psi_k \rho_k}{2\pi \parallel \boldsymbol{d} \|^2}, \tag{2}$$

and

$$P_{\text{IRS},k}(\boldsymbol{v}) = \int_{S_k} \frac{B_k R(\alpha_k, \beta_k)\cos\psi_k}{\|\boldsymbol{d}_{\boldsymbol{k},2}\|^2} \mathrm{d}S_k, \tag{3}$$

$$\begin{aligned} &R(\alpha_k, \beta_k) \\ &\quad = (2r_k \cos\beta_k + (1-r_k)(m_k+1)(\cos(\beta_k - \alpha_k))^{m_k}), \end{aligned} \tag{4}$$

$$B_k = \frac{P_0(m_k+1)A_{\text{PD}}(\cos\phi_k)^{m_k} \cos\alpha_k T_s(\psi_k) g\,(\psi_k)\,\rho_k}{4\pi^2 \parallel \boldsymbol{d}_{\boldsymbol{k},1} \|^2}, \tag{5}$$

where P_0 is the light source power, A_{PD} is the PD receiving area, ϕ_k is the light source irradiance angle, ψ_k is the PD incident angle, ρ_k is the reflection coefficient, $T_s(\psi_k)$ and $g(\psi_k)$ are the optical filter gain and optical concentrator gain of the PD, α_k is the incidence angle from LED to the IRS, β_k is the irradiance angle from the IRS to the PD, r_k represents the diffuse component coefficient, m_k is the Lambertian order, respectively.

B. DRSS Scheme for a Single LED Positioning

In traditional VLP systems, multiple light sources are typically required to achieve accurate positioning. However, when the LOS link from the light source is obstructed, the positioning performance degrades significantly or may even fail entirely. Additionally, a portion of the power received by the PD originates from non-line-of-sight (NLOS) channels, which are often regarded as interference. The introduction of IRS transforms this paradigm by not only utilizing the energy from NLOS channels but also enabling positioning even in the absence of an LOS link. The IRS enable high-precision positioning under a single light source by constructing NLOS links. These NLOS links can be regarded as coming from multiple virtual light sources. However, due to the complex interplay among received optical power, distance, and various angles during IRS reflection, directly determining the virtual light source's position is challenging. Therefore, this work uses the known positions of the light source and IRS to calculate the optical power received by the IRS and then regards the position of the IRS as the position of the virtual light source for positioning estimation.

During IRS positioning, a set of IRS units are turned on, and only two IRS units change the switch state at the same time in the process of adjusting the IRS, one from open to off, and one from off to on, and the optical power at the moment is received after the adjustment is completed. The above process is called adjusting IRS once. Based on formula (6), different receiving powers $P_R(\boldsymbol{v})$ can be obtained by adjusting the IRS multiple times. $P_{\text{LOS}}(\boldsymbol{v})$ and other IRS will not change when the PD position remains unchanged. The received power expression is

$$P_{\text{RX},i}(\boldsymbol{v}) = P_{\text{LOS}}(\boldsymbol{v}) + P_{\text{IRS},i}(\boldsymbol{v}) + \sum_{k=1}^{N} P_{\text{IRS},k}(\boldsymbol{v}) + \eta_i, \tag{6}$$

where i represents the i_{th} adjustment, and $P_{\text{IRS,i}}(\boldsymbol{v})$ is the IRS turned on in the ith adjustment. $P_{\text{LOS}}(\boldsymbol{v})$ is equal to 0 when the LOS link is blocked. This work subtracts the power from the power of the previous adjustment to get the power difference $D_{\text{i}}(\boldsymbol{v})$, that is

$$\begin{aligned} D_i(\boldsymbol{v}) &= P_{\text{RX},i}(\boldsymbol{v}) - P_{\text{RX},i-1}(\boldsymbol{v}) \\ &= P_{\text{IRS},i}(\boldsymbol{v}) - P_{\text{IRS},i-1}(\boldsymbol{v}) + e, \end{aligned} \tag{7}$$

where e is the error caused by channel noise. Since the difference of two independent Gaussian noises is still Gaussian, e is also Gaussian noise. For the convenience of subsequent analysis, let $r_k = 0$ for diffuse reflection mode, and different angles are represented by vectors:

$$\begin{cases} \cos\phi_k = \frac{\boldsymbol{n}_0 \cdot \boldsymbol{d}_{\boldsymbol{k},1}}{\|\boldsymbol{d}_{\boldsymbol{k},1}\|} \\ \cos\alpha_k = -\frac{\boldsymbol{n_k} \cdot \boldsymbol{d}_{\boldsymbol{k},1}}{\|\boldsymbol{d}_{\boldsymbol{k},1}\|} \\ \cos\beta_k = \frac{\boldsymbol{n_k} \cdot \boldsymbol{d}_{\boldsymbol{k},2}}{\|\boldsymbol{d}_{\boldsymbol{k},2}\|} \\ \cos\psi_k = -\frac{\boldsymbol{n_{\text{PD}}} \cdot \boldsymbol{d}_{\boldsymbol{k},2}}{\|\boldsymbol{d}_{\boldsymbol{k},2}\|} \end{cases}. \tag{8}$$

Since the IRS position and rotation angle are known, after multiple IRS adjustments, according to $\| \boldsymbol{d}_{\boldsymbol{i},2}\|^2 = (x-x_{\boldsymbol{i}})^2 + (y-y_{\boldsymbol{i}})^2 + (z-z_{\boldsymbol{i}})^2$, the simultaneous Eqs. (3), (4), (5), (7), and (8) can be used to obtain a system of equations about x, y, and z. Due to the high-order terms in the system of equations, the Levenberg–Marquardt algorithm is applied to solve the system of equations to achieve positioning:

$$\begin{cases} D_1(x,y,z) = P_{\mathrm{IRS},2}(x,y,z) - P_{\mathrm{IRS},1}(x,y,z) \\ D_2(x,y,z) = P_{\mathrm{IRS},3}(x,y,z) - P_{\mathrm{IRS},2}(x,y,z) \\ \cdots \\ D_{\mathrm{i}-1}(x,y,z) = P_{\mathrm{IRS,i}}(x,y,z) - P_{\mathrm{IRS,i}-1}(x,y,z) \end{cases}. \tag{9}$$

C. Positioning Error Evaluation

CRLB is used as the lower bound of the unbiased estimate variance to analyze the theoretical error. In the given conditions, the ideal minimum value of the positioning error can be obtained, which is independent of the positioning method and depends on the constructed noise model. CRLB is used to guide the optimized parameters. The following is the specific derivation process of CRLB.

Since the noise component in Eq. (1) is Gaussian noise and $P_{\mathrm{LOS}}(\boldsymbol{v})$ is equal to 0 when the LOS link is blocked, Eq. (1) can be rewritten as

$$P_{\mathrm{RX}}(\boldsymbol{v}) = \sum_{k=1}^{N} P_{\mathrm{RIRS},k}(\boldsymbol{v}) = \sum_{k=1}^{N} P_{\mathrm{IRS},k}(\boldsymbol{v}) + \eta, \tag{10}$$

where $P_{\mathrm{RIRS},k}(\boldsymbol{v})$ represents the actual power received from the k_{th} IRS. Since the noise of each channel is independent of each other, the log-likelihood function of $\boldsymbol{v}$ under the specified received power can be obtained as follows:

$$\begin{aligned} &\ln p(P_{\mathrm{RX}}(\boldsymbol{v})|\boldsymbol{v}) \\ &= -\sum_{k=1}^{N} \frac{1}{2{\sigma_k}^2}\left(P_{\mathrm{RIRS},k}(\boldsymbol{v}) - P_{\mathrm{IRS},k}(\boldsymbol{v})\right)^2 - J\ln\left(\sqrt{2\pi}\sigma_k\right), \end{aligned} \tag{11}$$

where σ_k^2 is the variance of the noise function η_k of the k_{th} IRS channel, and J is a constant. Based on Eq. (11), this work gets the first-order derivative and second-order derivative of the log-likelihood function:

$$\frac{\partial \ln p(P_{\mathrm{RX}}(\boldsymbol{v})|\boldsymbol{v})}{\partial v_{a_1}} = -\sum_{k=1}^{N} \frac{\eta_k}{{\sigma_k}^2}\left(\frac{\partial P_{\mathrm{RIRS},k}(\boldsymbol{v})}{\partial v_{a_1}} - \frac{\partial P_{\mathrm{IRS},k}(\boldsymbol{v})}{\partial v_{a_1}}\right), \tag{12}$$

where $\eta_k = P_{\mathrm{RIRS},k}(\boldsymbol{v}) - P_{\mathrm{IRS},k}(\boldsymbol{v})$. Since $P_{\mathrm{RIRS},k}(\boldsymbol{v})$ is the known received optical power, it can be regarded as a constant term, so its derivative is equal to 0:

$$\frac{\partial^2 \ln p(P_{\mathrm{RX}}(\boldsymbol{v})|\boldsymbol{v})}{\partial v_{a_1}\partial v_{a_2}} = \sum_{k=1}^{N} \frac{1}{{\sigma_k}^2}\left(\eta_k \frac{\partial^2 P_{\mathrm{IRS},k}(\boldsymbol{v})}{\partial v_{a_1}\partial v_{a_2}} - \frac{\partial P_{\mathrm{IRS},k}(\boldsymbol{v})}{\partial v_{a_1}}\frac{\partial P_{\mathrm{IRS},k}(\boldsymbol{v})}{\partial v_{a_2}}\right), \tag{13}$$

where $a_1, a_2 \in \{1,2,3\}$, v_{a_i} represents the a_ith variable of $\boldsymbol{v}$. The expected expression of the second-order derivative is as follows:

$$\mathbb{E}\left[\frac{\partial^2 \ln p(P_{\mathrm{RX}}(\boldsymbol{v})|\boldsymbol{v})}{\partial v_{a_1}\partial v_{a_2}}\right] = -\sum_{k=1}^{N} \frac{1}{{\sigma_k}^2}\frac{\partial P_{\mathrm{IRS},k}(\boldsymbol{v})}{\partial v_{a_1}}\frac{\partial P_{\mathrm{IRS},k}(\boldsymbol{v})}{\partial v_{a_2}}. \tag{14}$$

The expressions for each element in the Fisher information matrix (FIM) are

$$[\boldsymbol{I}(\boldsymbol{v})]_{a_1,a_2} = -\mathbb{E}\left[\frac{\partial^2 \ln p(P_{\mathrm{RX}}(\boldsymbol{v})|\boldsymbol{v})}{\partial v_{a_1}\partial v_{a_2}}\right]. \tag{15}$$

Therefore, combining Eqs. (9) and (10), the complete expression of FIM is as follows:

$$I(\boldsymbol{v}) = \sum_{k=1}^{N}\frac{1}{{\sigma_k}^2}\begin{pmatrix} \frac{\partial P_{\mathrm{IRS},k}}{\partial v_{a_1}}\frac{\partial P_{\mathrm{IRS},k}}{\partial v_{a_1}} & \frac{\partial P_{\mathrm{IRS},k}}{\partial v_{a_1}}\frac{\partial P_{\mathrm{IRS},k}}{\partial v_{a_2}} & \frac{\partial P_{\mathrm{IRS},k}}{\partial v_{a_1}}\frac{\partial P_{\mathrm{IRS},k}}{\partial v_{a_3}} \\ \frac{\partial P_{\mathrm{IRS},k}}{\partial v_{a_1}}\frac{\partial P_{\mathrm{IRS},k}}{\partial v_{a_2}} & \frac{\partial P_{\mathrm{IRS},k}}{\partial v_{a_2}}\frac{\partial P_{\mathrm{IRS},k}}{\partial v_{a_2}} & \frac{\partial P_{\mathrm{IRS},k}}{\partial v_{a_2}}\frac{\partial P_{\mathrm{IRS},k}}{\partial v_{a_3}} \\ \frac{\partial P_{\mathrm{IRS},k}}{\partial v_{a_1}}\frac{\partial P_{\mathrm{IRS},k}}{\partial v_{a_3}} & \frac{\partial P_{\mathrm{IRS},k}}{\partial v_{a_2}}\frac{\partial P_{\mathrm{IRS},k}}{\partial v_{a_3}} & \frac{\partial P_{\mathrm{IRS},k}}{\partial v_{a_3}}\frac{\partial P_{\mathrm{IRS},k}}{\partial v_{a_3}} \end{pmatrix}, \tag{16}$$

By combining Eqs. (3), (4), (5), (8), (11), (12), (13), (14), and (15), this work can get the specific expression of $\boldsymbol{I}(\boldsymbol{v})$, where the first derivatives of $P_{\mathrm{IRS,i}}(\boldsymbol{v})$ are as follows:

$$\begin{aligned} &\frac{\partial P_{\mathrm{IRS},k}(\boldsymbol{v})}{\partial x} \\ &= \int_{S_k} N_k\,(z_k - z)\left(\frac{-4(x-x_k)\boldsymbol{n}_{\boldsymbol{k}}\cdot \boldsymbol{d}_{\boldsymbol{k},2}}{\|\boldsymbol{d}_{\boldsymbol{k},2}\|^6} + \frac{n_{k,x}}{\|\boldsymbol{d}_{\boldsymbol{k},2}\|^4}\right)\mathrm{d}S_k, \end{aligned} \tag{17}$$

$$\begin{aligned} &\frac{\partial P_{\mathrm{IRS},k}(\boldsymbol{v})}{\partial y} \\ &= \int_{S_k} N_k(z_k - z)\left(\frac{-4(y-y_k)\boldsymbol{n}_{\boldsymbol{k}}\cdot \boldsymbol{d}_{\boldsymbol{k},2}}{\|\boldsymbol{d}_{\boldsymbol{k},2}\|^6} + \frac{n_{k,y}}{\|\boldsymbol{d}_{\boldsymbol{k},2}\|^4}\right)\mathrm{d}S_k, \end{aligned} \tag{18}$$

$$\begin{aligned} &\frac{\partial P_{\mathrm{IRS},k}(\boldsymbol{v})}{\partial z} \\ &= \int_{S_k} N_k\left((z_k - z)\left(\frac{-4(z-z_k)\boldsymbol{n}_{\boldsymbol{k}}\cdot \boldsymbol{d}_{\boldsymbol{k},2}}{\|\boldsymbol{d}_{\boldsymbol{k},2}\|^6} + \frac{n_{k,z}}{\|\boldsymbol{d}_{\boldsymbol{k},2}\|^4}\right)\right. \\ &\quad \left. - \frac{\boldsymbol{n}_{\boldsymbol{k}}\cdot \boldsymbol{d}_{\boldsymbol{k},2}}{\|\boldsymbol{d}_{\boldsymbol{k},2}\|^4}\right)\mathrm{d}S_k, \end{aligned} \tag{19}$$

where N_k is the optical power received by the k_{th} IRS, and S_k is the reflection area of the k_{th} IRS.

Therefore, the CRLB of each position can be calculated in given conditions. According to Ref. [19], if the matrix is invertible, then the lower bound of the mean square error of the estimated position $\boldsymbol{x}$, i.e., CRLB, has the following relationship:

$$\mathbb{E}[||\mathbf{e}||^2] \geq \mathrm{Tr}(\mathbf{I}(\boldsymbol{v})^{-1}) = \mathrm{CRLB}. \tag{20}$$

When the propagation model of IRS is known, this work can get the specific expression of CRLB. By analyzing and comparing CRLB in different situations, it can be regarded as a guide to continuously optimize related parameters.

3. SIMULATION RESULTS

A. CRLB for Different Orientation and Layout

In Sec. 2, this work derives the expression of the CRLB, which is regarded as a guide to analyze and optimize various parameters. The room with a size of $4\ \mathrm{m} \times 5\ \mathrm{m} \times 5\ \mathrm{m}$ is considered. This work evenly places the IRS on one wall of the room and at only one height. The simulation parameters are shown in Table 1.

Figure 2(a) shows the CRLB in the initial deployment, where the IRS are placed on the wall at $Y = 0$ m, with a height of $Z = 4$ m, and the X-coordinates range from 0 to 5, with 16 IRS in total. The IRS are facing inward by default. When Y is larger than 1 m, the farther away from the IRS, the larger its CRLB. The farther away from the IRS, the smaller its received power, and the greater the impact of noise. Meanwhile, it can be seen that in the area of Y less than 1 m, the closer to the IRS, the larger the CRLB. The reason is that the IRS are facing horizontally inward at this time. When the PD is close to the bottom of the IRS, the irradiance angles of IRS are close to 90 deg, so the power received by the PD will be very small, and it will be greatly affected by noise. To improve this situation, this work chooses to optimize the rotation angles of the IRS. By rotating at different angles, this work calculates the maximum power when reaching the PD, and then uses the rotation angles at this time to calculate the CRLB of the PD at each position, as shown in Fig. 2(b). It can be seen that the power is increased after angle optimization at the PD, which can achieve more accurate positioning. Additionally, due to the flexibility of the IRS angle rotation, it compensates for the increased error closer to the IRS observed in Fig. 2(a).

Under the condition of optimizing the IRS rotation angle for maximum power, this work also analyzes the impact of IRS quantity and layout on the CRLB. Figures 3(a) and 3(b) show the CRLB for 12 IRS and 100 IRS, respectively, with the same angle settings as in Fig. 2(b), corresponding to the optimal power angle. Compared to Fig. 2(b), it can be observed that as the quantity of IRS increases, the information received by the PD increases, leading to a reduction in the overall CRLB. However, it is also evident that the rate of CRLB decrease diminishes with the increasing quantity of IRS, indicating that the information provided by the IRS in the designated area has an upper limit. When the quantity of IRS reaches a certain threshold, the IRS have already provided most of the available information for the deployment area, and further increasing the quantity of IRS contributes little additional useful information. In fact, under high noise conditions, noise dominates the system performance, making the incremental benefits of adding more IRS beyond a certain threshold negligible. Therefore, by judiciously selecting the quantity of IRS and evenly distributing them, sufficient information can be provided while avoiding the waste of unnecessary IRS deployment.

Table 1. Simulation Parameters

Parameter	Value
Lambertian order m	1
LED transmit optical power P_0	6 W
Gain of the optical filter T_s	1
Open-loop voltage gain g	10
PD detector physical area A_{PD}	1 cm^2
LED coordinate	(2 m, 2.5 m, 5 m)

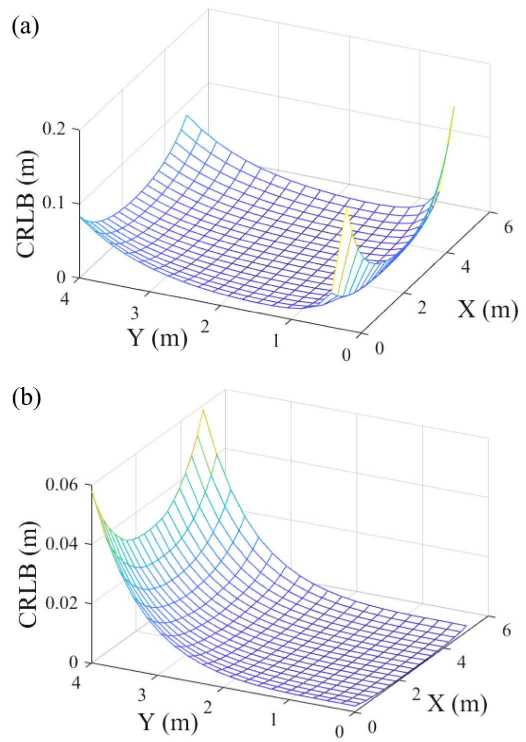

Fig. 2. CRLB under a single-wall deployment with different rotation angles. (a) Initial angle. (b) Optimized angle.

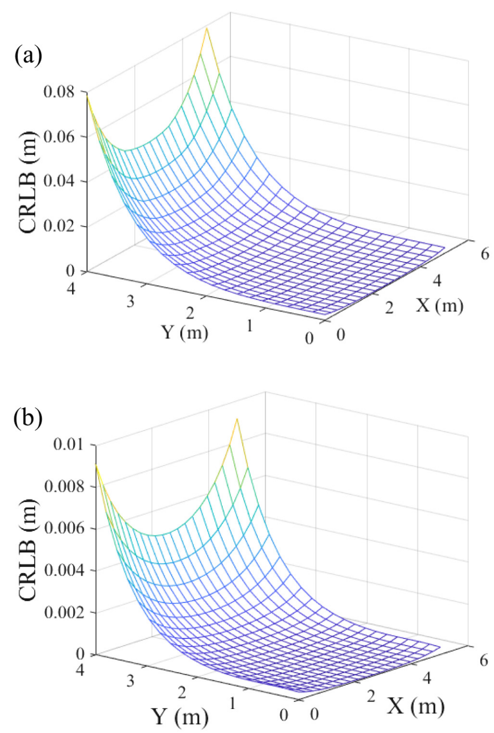

Fig. 3. CRLB under a single-wall with different quantity of IRS. (a) 12 IRS. (b) 100 IRS.

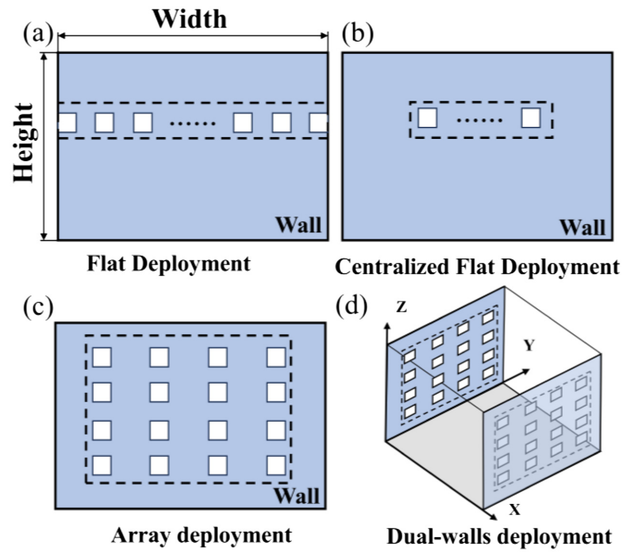

Fig. 4. Different IRS layouts. (a) Flat deployment. (b) Centralized flat deployment. (c) Array deployment. (d) Dual-walls deployment.

Considering the inherent information upper limit of the flat IRS deployment, this work further investigates the CRLB under different layout strategies. Figure 4 illustrates three layouts, labeled as flat deployment, centralized flat deployment, array deployment, and dual-walls deployment, with the dotted box indicating the designated area for IRS deployment.

In Fig. 2(b), the IRS are arranged uniformly at the same height in a flat deployment. Conversely, Fig. 5(a) adopts a more centralized deployment, with the X-coordinates of the IRS distributed between 1 m and 4 m. This concentrated deployment reduces the diversity of the positional information provided by the IRS, resulting in increased estimation errors. The errors are particularly pronounced at positions farther from the IRS, where the diminished coverage significantly affects performance. Figure 5(b) presents a 4×4 array deployment with the total quantity of IRS maintained at 16. Although the information contributed by IRS at the same height decreases, the introduction of IRS at different heights significantly enhances spatial diversity, providing richer and more comprehensive positional information. As a result, the CRLB achieved in the 4×4 array deployment is considerably lower than that observed in Fig. 2(b), demonstrating the effectiveness of diversified IRS layouts in improving positioning accuracy.

From the above analysis, it can be seen that the trend of the positioning error of a single wall IRS in the power optimum angle is directly related to the distance from PD to IRS. The farther the distance, the higher the CRLB. Therefore, when using IRS to achieve positioning, the area close to the IRS will have a better effect. Moreover, appropriately increasing the quantity of IRS and modifying their layout can further optimize positioning accuracy.

To compensate for the high errors observed at distant locations, IRS are symmetrically deployed on the opposite wall of the room, as shown in Fig. 4(d). This approach not only increases the total quantity of IRS but also incorporates additional positional information from the other wall, thereby enhancing the overall localization performance. Figures 6(a) and 6(b) are the flat deployment and array deployment in dual-walls, respectively, with 16 IRS on each wall. Since the light source is set at the top of the center of the room, the CRLB is symmetrically distributed when IRS are placed symmetrically. Meanwhile, with a dual-wall configuration, the IRS from the other wall provides more spatial information, and the error is much lower than that of a single wall. Moreover, the effect is still better when placed in an array. If conditions allow, deploying IRS on both walls can result in a significant improvement in positioning accuracy.

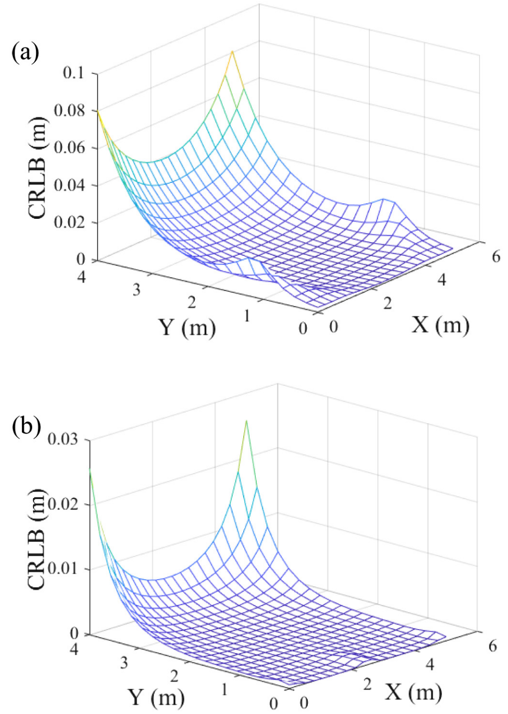

Fig. 5. CRLB under a single-wall in different layout. (a) Centralized flat deployment. (b) Array deployment.

From the perspective of quantity, increasing the quantity of IRS in a certain area will indeed improve positioning accuracy, but the amount of information in these areas is limited, so when the quantity reaches a certain value, the improvement will tend to saturate. From the perspective of layout, evenly placing IRS can obtain more regional positioning information and improve positioning accuracy, which is the same as the previous conclusion. Therefore, maximizing the acquisition of positioning information is the essence of improving positioning accuracy.

B. Positioning Results Using the DRSS Scheme

The above section mainly analyzes the impact of IRS' rotation angle, quantity, and layout on CRLB. Next, this work chooses to realize the VLP, which is placed on a single wall with 16 IRS. Figure 7(a) is a positioning scatter plot based on the DRSS scheme mentioned in Sec. 2. The actual position is the dot, where $Z = 0$, and the X and Y coordinates are evenly distributed between (0.5, 4.5) and (0.5, 3.5), respectively. The noise level is set to 110 dB, defined as $10\lg(1/\sigma^2)$, and 16 IRS are flatly deployed on a single wall. There are 5×5 test points in total. Each point is tested three times. The cross mark is the estimated

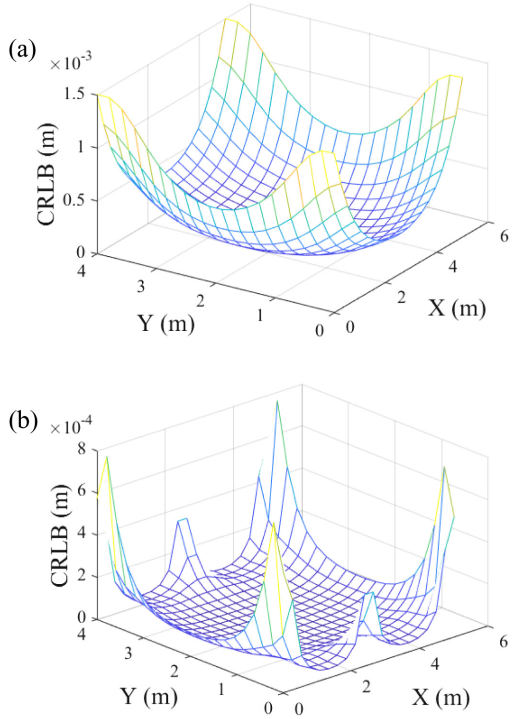

Fig. 6. CRLB under dual-walls deployment in different layout. (a) Flat deployment. (b) Array deployment.

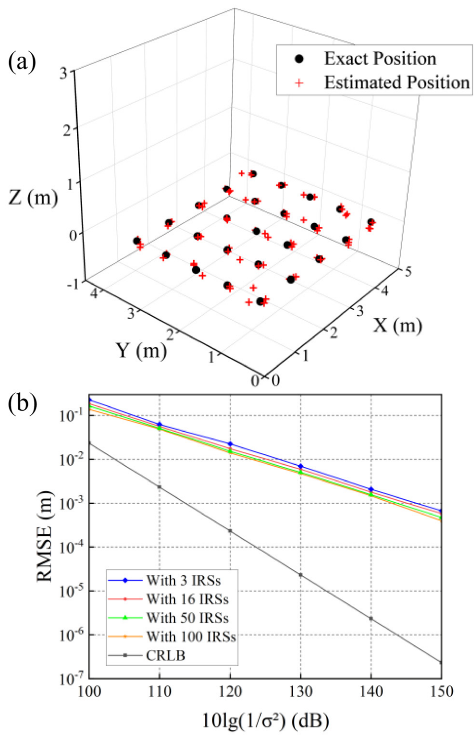

Fig. 7. Position estimation results. (a) Comparison of estimated and exact positions. (b) RMSE versus noise level.

position. Figure 7(b) shows the root mean square error (RMSE) at points (0.5, 3.5, 0) as it changes with noise. Since the IRS are set on a single wall, the RMSE of this point can be used as the upper bound of the room positioning error through the previous CRLB analysis, reflecting the specific positioning effect of the entire environment. As the noise decreases, it can be seen that the estimated RMSE is also decreasing like CRLB. At the same time, as the quantity of IRS increases, a reduction in RMSE is observed; however, the rate of improvement diminishes with further increases. For example, under a noise level of 110 dB, the positioning errors achieved with 3, 16, 50, and 100 IRS are 0.170 m, 0.056 m, 0.051 m, and 0.049 m, respectively. Notably, the improvement in positioning accuracy becomes marginal as the quantity of IRS increases from 50 to 100, indicating that the additional IRS contribute only minimal new information to the system. This suggests that the positional information provided by the IRS approaches saturation, where further increases in IRS quantity result in diminishing reductions in positioning error. Since the IRS in this scenario are in flat deployment, the effective information provided approaches saturation when the quantity of IRS reaches a certain threshold. This aligns with the analysis of the CRLB discussed earlier.

To achieve better positioning performance, this work also investigates the positioning performance in different IRS layouts. Figure 8(a) shows the scatter plot based on the dual-walls deployment, which exhibits significantly improved positioning accuracy, reducing the overall error to 0.014 m. Figure 8(b) illustrates the variation of RMSE with noise for the point located at (0.5, 3.5, 0) in different layout. By adjusting the IRS layout, richer positioning information can be provided, enabling more precise positioning. For instance, under a noise level of 110 dB, the positioning error under the dual-walls deployment is reduced from 0.056 to 0.014 m, which represents a substantial improvement compared to the errors observed under the single-wall flat deployment and single-wall array deployment, measured at 0.056 and 0.028 m, respectively.

The results show that the positioning accuracy of array placement is better than that of flat placement, and dual-walls array placement is superior to single-wall array placement. Compared to simply increasing the quantity of IRS, optimizing the layout is more effective in enhancing accuracy. This result is consistent with the conclusion of the previous theoretical analysis and further verifies the correctness of CRLB as a theoretical guide.

C. Analysis of PD Orientation

In practical positioning scenarios, it is challenging to ensure that the PD remains perfectly horizontal in deployment. Therefore, this work further analyzes the impact of PD orientation changes on positioning errors. Figure 9(a) is a positioning scatter plot when the PD is tilted 10 deg. In this scenario, the positioning error increases significantly from 0.056 to 0.591 m. Compared with Fig. 7(a), the positioning effect is significantly worse. Figure 9(b) shows the change of RMSE with noise when the PD is tilted 5 deg and 10 deg, respectively. When the noise level is set to 110 dB, the positioning errors for PD tilt angles of 5 deg and 10 deg are 0.281 and 0.591 m, respectively. This is because the positioning method is highly dependent on the received optical power, making the positioning performance sensitive to the tilt of the PDs. Due to the inherent error caused by the tilt, the noise in the transmission process can no longer improve the positioning effect when it is reduced to a certain extent, so further research is needed in terms of anti-tilt. By the way, in the

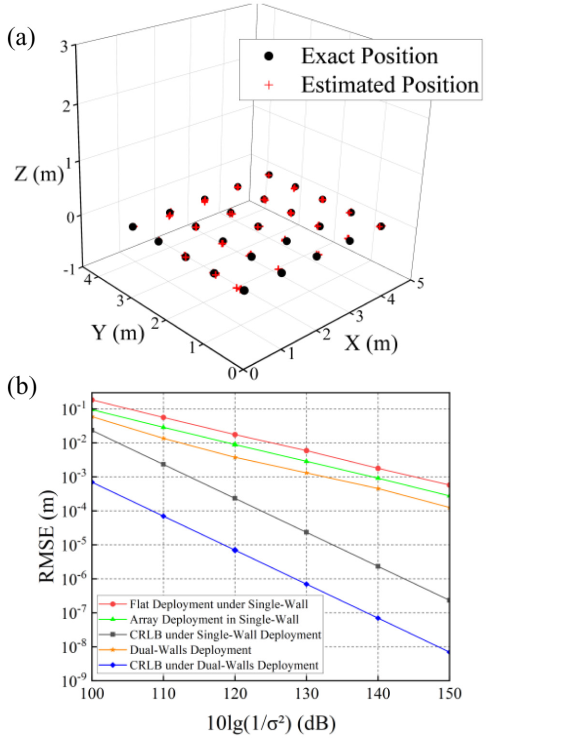

Fig. 8. Position estimation results. (a) Comparison of estimated and exact positions. (b) RMSE versus noise level in different deployment.

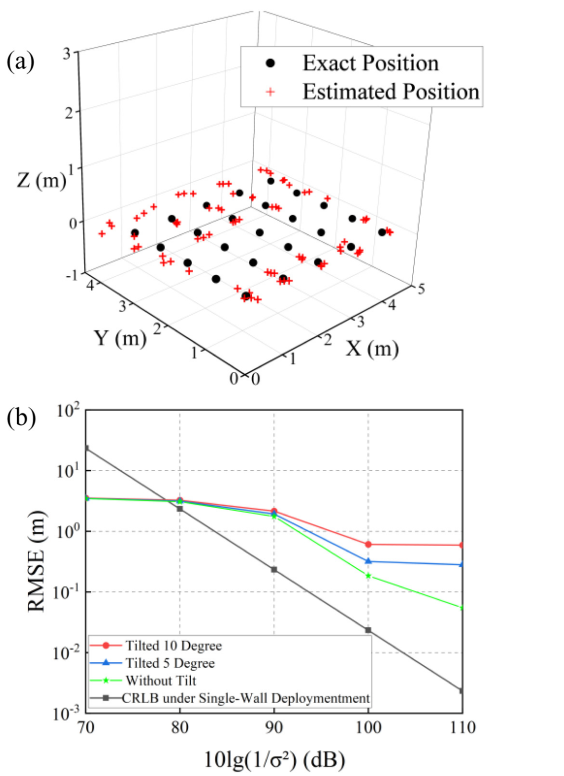

Fig. 9. Position estimation results. (a) Comparison of estimated and exact positions. (b) RMSE versus noise level at different tilt angles.

case of high noise, the CRLB is even higher than the estimated error. This is because the possible position of the PD in the room is limited to the room. The solution using the Levenberg–Marquardt algorithm is actually searching from various points in this room, and even in the worst case, the positioning error is only the size of the room. However, CRLB is not constrained by this room, so the RMSE of the estimated position is less than CRLB when there is large noise [20].

4. CONCLUSION

This paper presents a novel VLP scheme that utilizes single-light-source and IRS to address the limitations of traditional multi-light-source systems. By leveraging IRS to enhance and utilize NLOS signals, the DRSS scheme is proposed and achieves robust positioning even in scenarios in the absence of LOS. The study derives the CRLB to analyze the impact of key IRS parameters, including rotation angles, layouts, and quantities, on positioning accuracy. Simulation results validate the feasibility of the proposed scheme, achieving a positioning accuracy of 0.056 m with 16 IRS units under a noise level of 110 dB. Meanwhile, an optimized layout reduces the positioning error to 0.028 m, highlighting its importance in maximizing system performance. Moreover, the results reveal that while increasing the quantity of IRS reduces positioning error, the improvement diminishes beyond a certain threshold, emphasizing the significance of layout optimization over the mere increase in quantity. Additionally, the study on receiver tilt necessitates further research for practical scenarios. This work provides a valuable reference for advancing the integration of IRS technology into VLP systems.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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